

RECYCLN

Kawasaki-Style Pitch

1. PROBLEM / OPPORTUNITY

Waste collection is inefficient because waste operators often do not have accurate, real-time information about **where waste is accumulating, what type of waste is present, and which locations should be prioritized for collection.**

Unidentified dumpsites, illegal dumping, overflowing collection points, and changing waste volumes make manual inspection expensive and difficult to scale.

This creates an opportunity to use **satellite intelligence and AI** to detect potential waste sites, classify waste patterns, monitor changes, and provide collection operators with actionable information.

2. VALUE / PROPOSITION

RECYCLN is an AI-powered waste intelligence and collection platform that uses satellite and geospatial intelligence to identify, map, classify, and monitor waste accumulation.

RECYCLN helps waste management companies and organizations:

- Detect potential waste accumulation sites using satellite imagery.
- Use AI to classify and analyze identified waste locations.
- Monitor changes in waste sites over time.
- Prioritize locations requiring collection or intervention.
- Plan more efficient collection routes.
- Track collection activities and outcomes.
- Connect recovered materials with recycling and processing facilities.

Our value is simple:

“Know where the waste is. Understand what it is. Collect it efficiently. Recover its value.”

3. TECH STACK

Satellite & Geospatial Intelligence

• Sentinel-1 / Sentinel-2 • Landsat • OpenStreetMap • PostGIS • MapLibre / CesiumJS

AI & Machine Learning

• Computer Vision • Image classification and object detection • Change detection • Predictive analytics • Large Language Models • Python • PyTorch / Scikit-learn

Platform

- React + Vite • Tailwind CSS • React Native + Expo • Node.js + TypeScript • NestJS

Data Infrastructure

- PostgreSQL + PostGIS • Redis • Object Storage • pgvector

Optimization

- OR-Tools • Geospatial routing • AI-powered collection and logistics optimization

4. BUSINESS MODEL

RECYCLN will operate primarily as a **B2B Software-as-a-Service platform**.

Rather than building and maintaining all software infrastructure internally, we will **outsource specialized software development and technical implementation** while maintaining ownership of the RECYCLN platform, product strategy, data architecture, AI models, and business relationships.

Revenue will come from:

- **Software subscriptions** — waste management companies and organizations pay monthly or annually for access to the platform.
- **Enterprise contracts** — customized deployments for large waste operators and organizations.
- **Geospatial intelligence services** — organizations pay for waste-site detection, monitoring, and geographic analysis.
- **AI analytics** — premium access to forecasting, classification, prioritization, and operational intelligence.
- **API access** — businesses can integrate RECYCLN's waste intelligence into their existing systems.
- **Implementation and integration fees** — charges for connecting RECYCLN with existing fleet, ERP, GIS, or waste-management systems.

The business therefore scales through **software and intelligence**, without requiring RECYCLN to own waste trucks, recycling facilities, or physical collection infrastructure.

THE BUSINESS IN ONE SENTENCE

RECYCLN uses satellite intelligence and AI to identify where waste is accumulating, understand what is there, and help waste operators collect and recover it more efficiently.